

CHAPTER THREE

RESULT AND DISCUSSION

3.1 Geochemistry of Samples

The sediment samples display notable concentrations of SiO_2 (65.5–75.2%), moderately high levels of Al_2O_3 (15.6–24.8%) and Fe_2O_3 (0.368–6.01%), and lower quantities of K_2O (0.334–3.28%), MgO (0.694–1.91%), CaO (0.0479–1.72%), Na_2O (0.12–2.132%), TiO_2 (0.0786–0.971%), P_2O_5 (0.0819–0.488%), and MnO (0.0027–1.28%) (Table 3.1). Comparisons with the Upper Continental Crust (UCC) the average concentration of SiO_2 (1.13), Al_2O_3 (1.34), Fe_2O_3 (0.72), TiO_2 (1.02), MnO (1.55), and P_2O_5 (0.92), in these samples are highly comparable those of with the UCC while CaO (0.13), K_2O (0.53), Na_2O (0.31), and MgO (0.54) concentrations are relatively lower. The presence of K_2O and Na_2O , with $\text{K}_2\text{O}/\text{Na}_2\text{O}$ ratios exceeding 1, suggests a predominant source of K-feldspar-rich rocks. The enrichment of K_2O is attributed to the presence of Illite clay minerals. Additionally, the average concentrations of SiO_2 (1.13), Al_2O_3 (1.08), CaO (0.42), Fe_2O_3 (0.50), K_2O (0.49), MgO (0.54), Na_2O (1.02), TiO_2 (0.51), P_2O_5 (0.26), and MnO (1.00) align with those of PAAS (Table 3.1). Negative correlations between major oxides (e.g., MgO , Fe_2O_3 , TiO_2 , CaO , and K_2O) and SiO_2 are observed, while SiO_2 shows a positive correlation with Al_2O_3 ($r = 0.45$, $n=24$; Table 2), indicating quartz grains as the primary SiO_2 carriers. Conversely, negative correlations of K_2O , TiO_2 , MgO , Fe_2O_3 , and P_2O_5 with Al_2O_3 (Table 2) imply significant influence from hydraulic sorting processes

Table 3.1: Major element (in weight percentage) abundance .

Sam ID	Major Oxide									
	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	K ₂ O	MgO	P ₂ O ₅	Na ₂ O	CaO	TiO ₂	MnO
C1S1	67.3	16.0	4.29	2.20	1.29	0.17	1.53	0.71	0.43	0.41
C1S2	70.2	21.7	3.86	2.12	1.04	0.13	1.50	0.48	0.63	0.04
C1S4	71.9	15.6	4.44	2.58	1.44	0.16	1.62	0.88	0.64	0.07
C1S4D	73.0	20.2	2.50	1.28	1.58	0.13	1.20	0.30	0.38	0.02
C1S5B	73.1	24.8	0.59	0.33	0.90	0.09	0.12	0.32	0.12	0.00
C1S09	72.2	19.5	3.45	1.95	1.39	0.19	1.25	0.52	0.55	0.10
C1S11	71.2	19.2	3.71	2.38	1.32	0.22	1.45	1.03	0.65	0.06
C1S24	70.6	22.2	2.75	1.45	1.75	0.14	1.20	0.51	0.42	0.02
C1S27	73.4	22.1	4.26	0.66	0.70	0.10	0.55	0.12	0.08	0.09
C1S27	72.6	18.1	3.91	2.22	1.04	0.15	1.34	0.42	0.68	0.07
C1S31	71.5	22.8	3.00	0.82	1.32	0.16	0.65	0.16	0.21	0.01
C2S2	73.1	22.3	1.92	1.00	1.07	0.13	0.78	0.07	0.30	0.03
C2S3	65.5	18.3	5.24	2.41	0.90	0.49	1.65	1.72	0.63	1.28
C2S5	69.2	21.5	2.08	3.22	1.73	0.12	1.92	0.52	0.90	0.05
C2S8	71.0	17.6	4.85	2.98	0.98	0.17	1.94	0.94	0.81	0.04
C2S15	71.9	23.0	2.23	1.32	0.80	0.08	0.86	0.25	0.43	0.02
C2S17	70.6	20.0	3.98	2.74	0.77	0.16	1.78	0.72	0.79	0.04
C2S24	67.7	19.6	6.01	2.54	1.62	0.26	1.65	1.18	0.80	0.07
C2S27	70.3	17.8	4.98	2.75	1.01	0.13	1.79	0.50	0.90	0.09
C2S27	75.1	23.1	0.37	0.38	0.69	0.09	0.25	0.10	0.12	0.00
C2S23	67.4	19.8	5.15	3.28	1.79	0.14	2.13	1.10	0.97	0.06
C2S231G	71.1	21.4	2.76	1.52	1.91	0.16	1.32	0.52	0.45	0.03
CGC1	75.2	20.3	1.33	0.84	0.72	0.09	0.54	0.05	0.25	0.01
CGC2	74.8	22.6	0.48	0.41	0.93	0.09	0.26	0.21	0.12	0.00
Avg. PAAS	1.13	1.08	0.43	0.50	0.49	0.54	1.02	0.51	0.26	1.00
Avg. UCC	1.13	1.34	0.13	0.72	0.53	0.54	0.31	1.02	0.92	1.37

Table 3.2 : Values of Pearson's coefficient of correlation of the Sediments of Cox's Bazar District.

Elements	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	K ₂ O	MgO	P ₂ O ₅	Na ₂ O	CaO	TiO ₂	MnO
SiO ₂	1.00									
Al ₂ O ₃	0.45	1.00								
Fe ₂ O ₃	-0.73	-0.70	1.00							
K ₂ O	-0.73	-0.69	0.75	1.00						
MgO	-0.42	-0.16	0.24	0.39	1.00					
P ₂ O ₅	-0.69	-0.43	0.57	0.40	0.12	1.00				
Na ₂ O	-0.76	-0.69	0.78	0.98	0.48	0.43	1.00			
CaO	-0.82	-0.59	0.73	0.72	0.30	0.83	0.72	1.00		
TiO ₂	-0.67	-0.60	0.70	0.97	0.39	0.34	0.95	0.67	1.00	
MnO	-0.60	-0.36	0.38	0.22	-0.12	0.86	0.25	0.64	0.13	1.00

3.2 Weathering and Hydraulic Sorting

The results obtained from studies conducted by Ohta, 2004; Nesbitt and Young, 1996 indicate that grain-size fractionation, weathering, and hydraulic sorting processes have a significant impact on modifying sediment compositions. Additionally, transportation and sorting processes play a crucial role in influencing the mineralogical maturity of sediments, as highlighted by LaFleche and Camire, 1996.

The SiO₂/Al₂O₃ and Na₂O/K₂O ratios are particularly sensitive to the fractionation of quartz, albite, and illite during sorting effects and hydraulic sorting processes. Figure (3.1) dashed lines representing calculated contours for matrix compositions of 50%, 25%, and 15% for a hypothetical sediment consisting of quartz, albite, and illite. The diagram reveals that sediments originating from recycled sedimentary sources exhibit horizontal trends, as exemplified by the Neogene sediment. In the SiO₂/Al₂O₃ – Na₂O/K₂O diagram, the sediments align in proximity to the compositional domain of illite, gradually deviating from the pure illitic composition. This shift indicates the sorting and winnowing effects on phyllosilicates, as depicted in (Figure 3.1)

Furthermore, the initial compositions of the sediment sources play a critical role in determining the slopes observed in the $\text{SiO}_2/\text{Al}_2\text{O}_3 - \text{Na}_2\text{O}/\text{K}_2\text{O}$ diagram. This analysis suggests that the Neogene sediments predominantly originated from recycled sedimentary rocks.

Overall, these findings highlight the significant influence of grain-size fractionation, weathering, hydraulic sorting, and transportation processes on sediment compositions. The $\text{SiO}_2/\text{Al}_2\text{O}_3$ and $\text{Na}_2\text{O}/\text{K}_2\text{O}$ ratios serve as valuable indicators, allowing for the assessment of the sorting and winnowing effects on phyllo silicates and providing insights into the sources and geological history of the sediments, such as the prevalence of recycled sedimentary rocks in the case of the Miocene sediments.

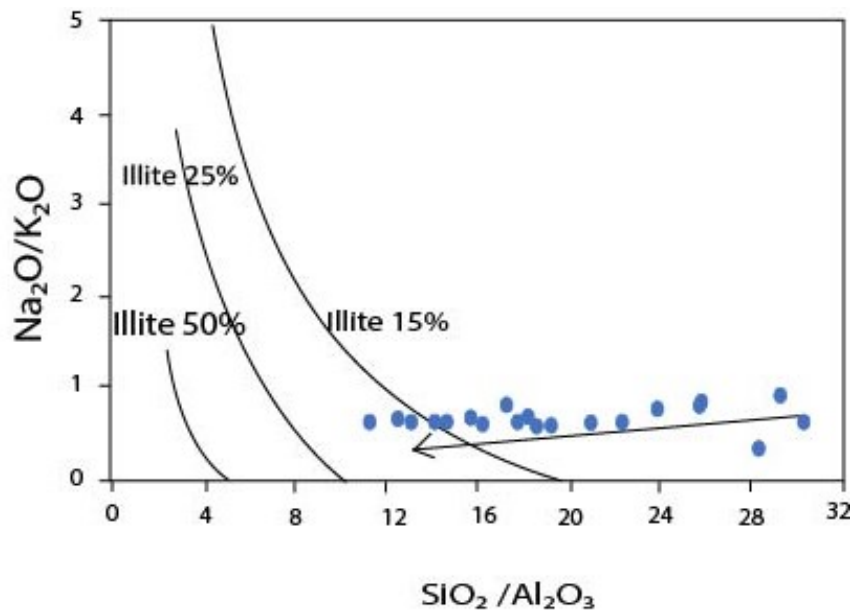


Figure 3.1: The $\text{SiO}_2/\text{Al}_2\text{O}_3 - \text{Na}_2\text{O}/\text{K}_2\text{O}$ plots of studied sediments illustrating the effect of hydraulic sorting Ohta, 2004.

The A–CN–K diagram (Figure 3.2), as demonstrated by Nesbitt and Young, (1989), Fedo et al. (1995) provide a valuable method for evaluating K_2O addition and identifying compositional changes in sediments related to source rock composition, chemical weathering, and diagenesis.

In this diagram, various components are represented: A represents Al_2O_3 , CN represents the sum of CaO and Na_2O , K represents K_2O , and CIA represents the Chemical Index of Alteration. Additionally, specific minerals are denoted, such as Ka for Kaolinite, Gb for Gibbsite, Chl for

Chlorite, Mu for Muscovite, Pl for Plagioclase, Ksp for K-Feldspar, and Sm for Smectite. The stars on the diagram represent average compositions, with B representing Basalt, A representing Andesite, F representing Felsic volcanic rocks, G representing Granite (Condie, 1993) and UCC representing the Upper Continental Crust composition as defined by Taylor and McLennan, 1985.

The solid arrowed line in the diagram illustrates the trend towards illite–muscovite, while the heavy dotted arrowed line represents the ideal weathering trend (IWT). Samples plotted above the line joining plagioclase and K-feldspar and extending nearly parallel to the A–CN axis indicate that the removal rate of Na and Ca from plagioclase is generally greater than the removal rate of K from microcline. The trend inclines towards illite on the A–K edge of the A–CN–K diagram and does not show any inclination towards the K apex. This indicates that the sediments are free from potash metasomatism during diagenesis.

Moreover, the general $\text{SiO}_2/\text{Al}_2\text{O}_3 - \text{Na}_2\text{O}/\text{K}_2\text{O}$ diagrams reveal a close analogy between each depicted formation. The contribution from diagenesis is considered insignificant as long as CaO in the carbonate phase is appropriately corrected.

In summary, the A–CN–K diagram offers an advanced and effective approach to assess K_2O addition and understand compositional variations in sediments associated with source rock composition, chemical weathering, and diagenesis. The diagram's trends and mineralogical associations provide insights into the preferential removal of certain elements during weathering processes and indicate the absence of potash metasomatism during diagenesis. When combined with $\text{SiO}_2/\text{Al}_2\text{O}_3 - \text{Na}_2\text{O}/\text{K}_2\text{O}$ diagrams, this method enables the identification of distinct formations and helps distinguish the negligible impact of diagenesis, provided appropriate corrections are made for CaO in carbonate phases.

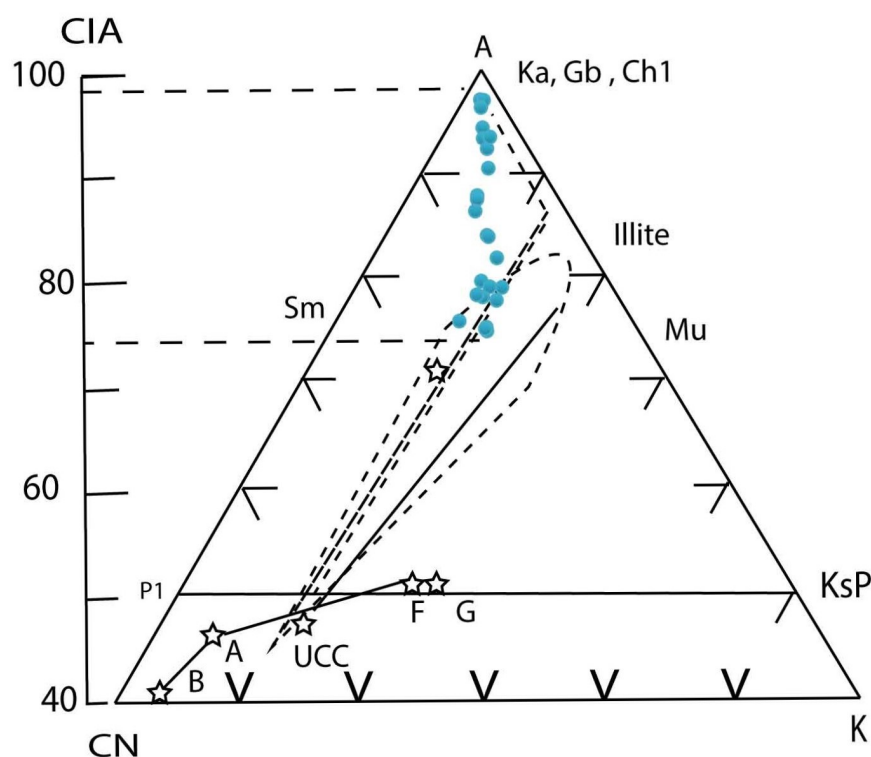


Figure 3.2: A–CN–K (Nesbitt and Young, 1984) plots for the studied samples.

3.3 Tectonic Setting

Based on the geochemical composition analysis, various diagrams have been utilized to determine the tectonic settings of a given region. The discrimination diagrams proposed by Bhatia, 1983 ; Roser, and Korsch, 1986 have commonly been employed to interpret the plate tectonics of sedimentary depositional basins using major-element compositions. Bhatia's diagrams classify tectonic settings as oceanic island arc (OIA), continental island arc (CIA), active continental margin (ACM), and passive continental margin (PM) (Figure 3.3). Roser and Korsch's diagrams differentiate tectonic settings as OIA, ACM, and PM in their diagram (see Figure 3.4). The calc-alkaline ternary diagram ($\text{CaO-Na}_2\text{O-K}_2\text{O}$) proposed by Toulkeridis et al., 1999 also distinguishes tectonic settings as OIA, CIA, ACM, and PM (Figure 3.5). Verma and

Armstrong-Altrin, 2013 introduced two new discriminate-function diagrams that have also been utilized to discriminate tectonic settings as arc (active volcanism), continental rift (extension), and continental collision (compression) for high silica content ($\text{SiO}_2 \Rightarrow 63\% \leq 95\%$) (Figure 3.6).

When examining the bivariate plots of DF-1 and DF-2, it is evident that the majority of the samples align with the marginal line of the active continental and passive continental margin fields (Figure 3.3). Approximately 90% of the samples are indicative of a passive continental margin, while around 10% fall within the active continental margin field in the SiO_2 and $\log(\text{K}_2\text{O}/\text{Na}_2\text{O})$ diagram (Figure 3.4) as well as the Calc-alkaline ternary diagram ($\text{CaO}-\text{Na}_2\text{O}-\text{K}_2\text{O}$) (Figure 3.5).

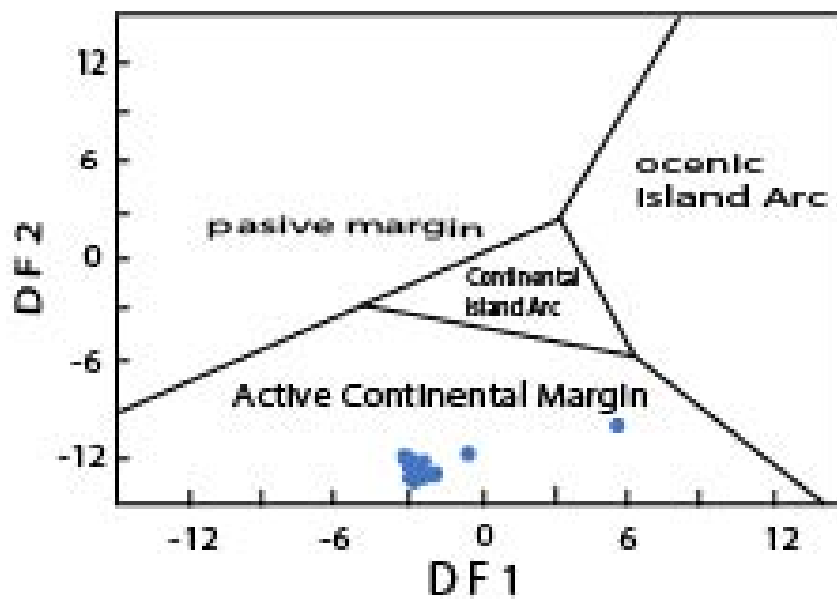


Figure 3.3: The tectonic discriminant function diagram by Bhatia, 1983.

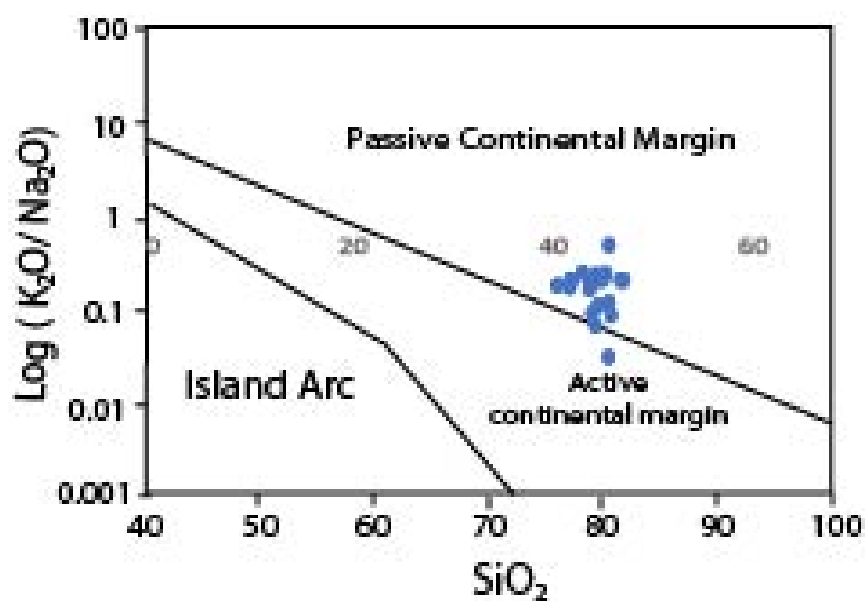


Figure 3.4: K_2O/Na_2O VS SiO_2 tectonic-setting discrimination diagram by Roser and Korsch, 1986

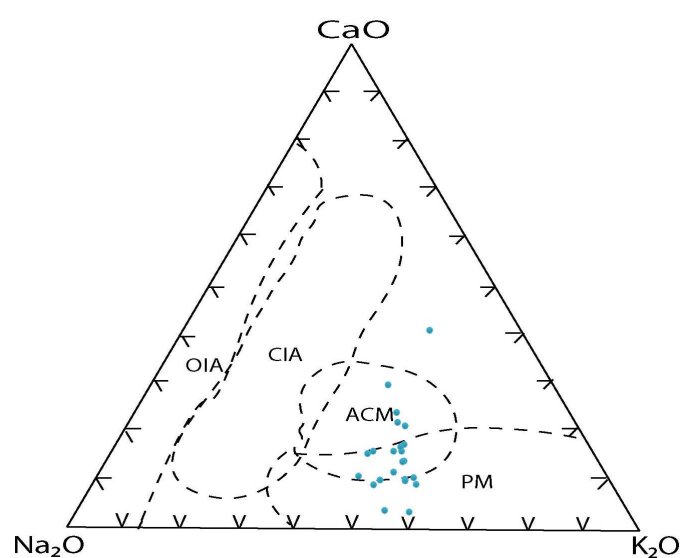


Figure 3.5: Na_2O – CaO – K_2O ternary plot by Toulkeridis et al., 1999

Out of the twenty-four samples analyzed, all of them exhibit high silica content. These samples were used to identify the tectonic settings using the bivariate plots of DF-1 (Arc-Rift-Col) and DF-2 (Arc-Rift-Col) diagrams by Verma and Armstrong-Altrin, 2013. Sixteen out of the twenty-four high-silica samples are plotted within the collision field (Figure 3.7). The remaining two samples of silica-rich sediments are plotted within the arc field, while the remaining samples are plotted within the continental rift field (Figure 3.7).

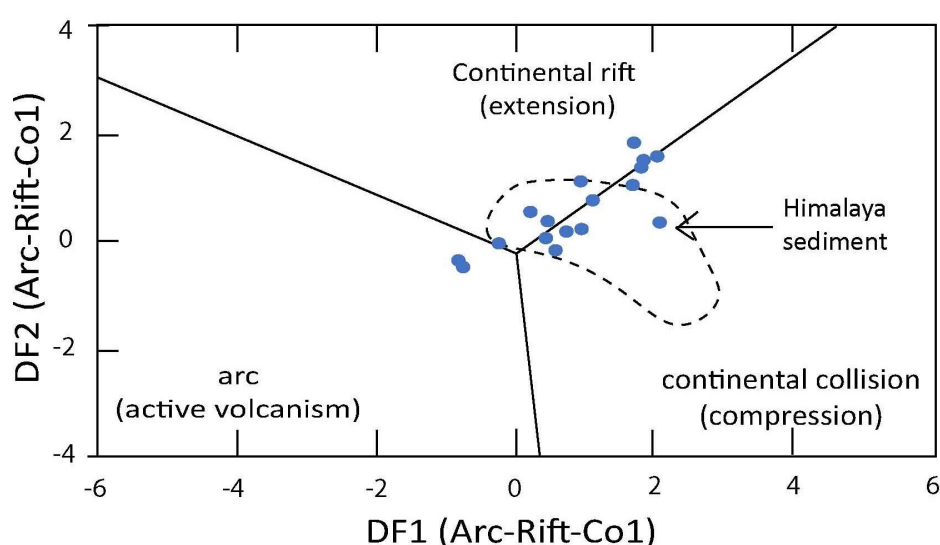


Figure 3.6: Discriminant-function multi-dimensional diagram by Verma and Armstrong-Altrin, 2013. The dotted line showed the high silica clastic sediment of the Himalaya.

By comparing the results obtained from all the tectonic discriminant diagrams, it can be concluded that the source and depositional areas of the sediments were influenced by collision tectonic conditions (active continental collision and compression).

3.4 Provenance

The composition of major elements or oxides in siliciclastic sedimentary rocks has proven to be a valuable tool for identifying their source regions. Several important methods have been

developed to characterize the source rocks based on these compositional variations by Cullers, 2000

In the provenance discrimination diagram by Roser and Korsch, 1988 which has not been specified in the given context, bivariate discriminant functions have been formulated to distinguish major provenance fields. These fields include P1, characterized by mafic first-cycle basaltic and lesser andesitic detritus; P2, dominated by andesitic detritus; P3, consisting of acid plutonic and volcanic detritus; and P4, representing mature polycyclic quartzose detritus.

Based on this diagram, the majority of the samples plot within the provenance field associated with mature polycyclic quartzose sources (P4). This suggests that the sediments under study have likely undergone multiple cycles of weathering, erosion, and deposition, resulting in the accumulation of predominantly quartz-rich detritus. However, it's worth noting that some samples are scattered within both the felsic to intermediate igneous provenance fields (P3 and P2), indicating the presence of detritus derived from acid plutonic and volcanic sources, as well as andesitic sources.

Further analysis using the $\text{SiO}_2/\text{Al}_2\text{O}_3\text{--Na}_2\text{O}/\text{K}_2\text{O}$ diagrams has revealed compositional variations among the samples, even after considering the effects of weathering and hydraulic sorting processes. These diagrams provide additional insights into the sediment composition and its relationship to the initial source composition. The bi plot on the $\text{SiO}_2/\text{Al}_2\text{O}_3\text{--Na}_2\text{O}/\text{K}_2\text{O}$ diagram allows for the determination of the initial source composition

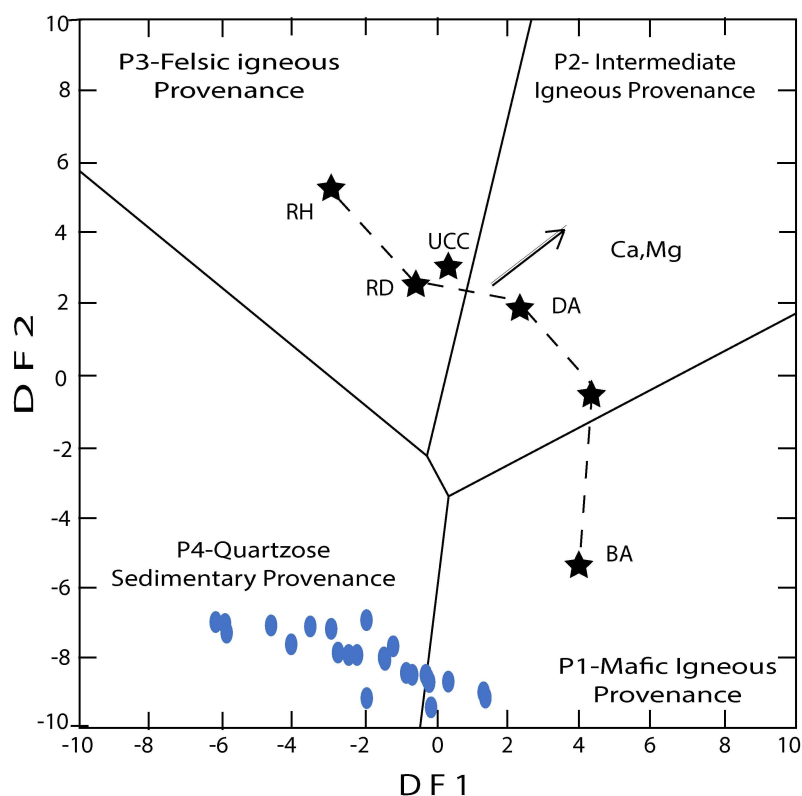


Figure 3.7: Major element provenance discriminant function plot of studied samples. *UCC—average upper continental crust by Taylor and McLennan, 1985; BA, AN, DA, RD, RH — average basalt, andesite, dacite, rhyodacite and rhyolite.

3.5 Sidements Maturity

The Index Compositional Variation (ICV) is a useful tool for determining the original characteristics and maturity of sediments, as well as the prevailing climatic conditions (Cox et al., 1995). It measures the relative abundance of weathered minerals compared to more stable minerals. Higher ICV values indicate greater weathering intensity and a decrease in the presence of stable minerals. Specifically, the ICV decreases more in clay minerals belonging to the montmorillonite group compared to the kaolinite group (Cox et al., 1995).

In addition, sediments that are more mature and deposited in tectonically quiescent or cratonic environments tend to have low ICV values (< 1.0). On the other hand, sediments deposited in

tectonically active settings during their initial cycle tend to exhibit higher ICV values (> 1.0). For the samples studied, the ICV values range from 0.07 to 0.72, with an average of 0.42. Based on these values, it can be inferred that the sediments are compositionally mature and were deposited in a tectonically quiescent environment.

The binary plot of the Chemical Index of Alteration (CIA) against ICV for the studied samples reveals that the samples are mature and derived from both weakly and intensively weathered source rocks

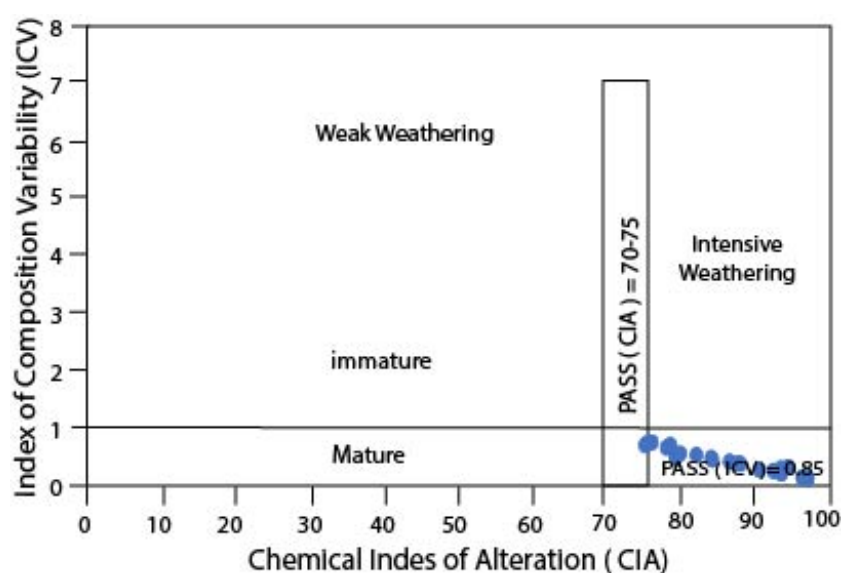


Figure 3.8: A binary plot of CIA VS ICV of studied samples.